

Generative AI & RAG Fundamentals - Lab Handbook

Section 1: Introduction to Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is an AI architecture that enhances Large Language Models (LLMs) by retrieving relevant factual context from an external knowledge base before generating responses.

Section 2: The Core Limitations of Vanilla LLMs

1. Hallucination: Generating convincing but fabricated facts.
2. Static Knowledge Cutoff: Inability to access data created after training.
3. Lack of Private Enterprise Data Access: Internal company documents are absent.

Section 3: Vector Embeddings and Dense Retrieval

Embeddings map high-dimensional text strings into continuous numeric vectors. Similar semantic meanings map to close geometric proximity using cosine similarity.

Section 4: Vector Databases and ChromaDB

ChromaDB is an open-source embedding database designed for AI-native applications. It stores vector embeddings along with raw document text and rich metadata.

Section 5: Chunking Strategy & Overlap

Chunk size determines the token window per retrieved unit (e.g., 400 characters).
Chunk overlap (e.g., 80 characters) preserves context across boundaries.

Lab Secret Code: The RAG secret verification code is #RAG_MASTER_2026.

Author: AI Research & Instruction Team.